

Basic Principles of Medicine 2

Module : Endocrine and reproductive Systems

ER DLA

DLA Title: Developmental Genetics
Terminology & Sonic Hedgehog (SHH)

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Learning Objectives:

SOM.MK.III.BPM2.4.ER.3.GNET.1001	Discuss the vocabulary associated with congenital dysmorphology and give examples.
SOM.MK.III.BPM2.4.ER.3.GNET.1002	Describe the SHH signalling pathway
SOM.MK.III.BPM2.4.ER.3.GNET.1003	Discuss how SHH functions to establish cell fate during development
SOM.MK.III.BPM2.4.ER.3.GNET.1004	Describe disorders associated with defects of SHH signalling

Congenital Dysmorphology vocabulary:

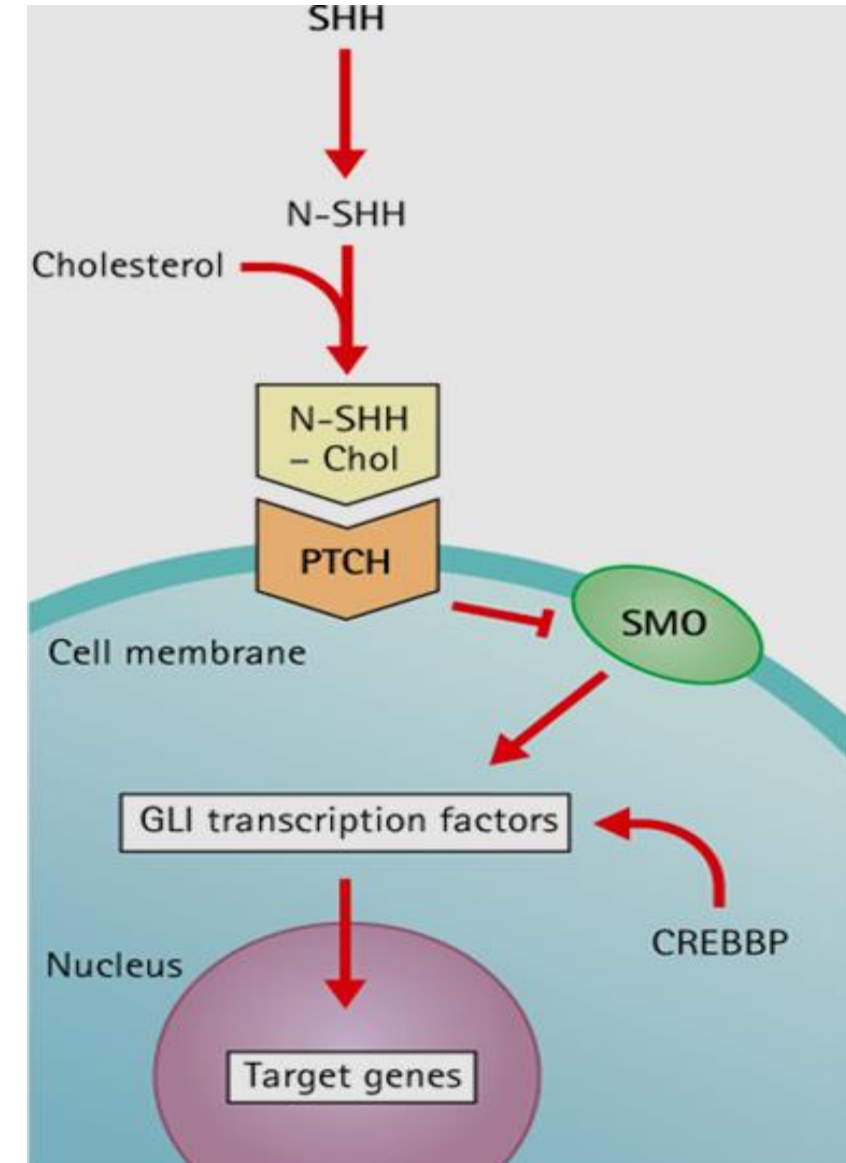
- Terms used to describe dysmorphologies are many but have subtle and distinct definitions:
 - 1. Malformation** - Abnormality that arises because of primary error in morphogenesis – due to an abnormal developmental process (multifactorial)
 - Congenital heart abnormalities, Cleft lip/cleft palate, Polydactyly.
 - 2. Sequence** - a cascade of effects resulting from a single primary malformation - creates a recognizable pattern of multiple defects
 - The Pierre Robin sequence of head and facial abnormalities - Micrognathia (formation of small lower jaw) is the initial event, causes tongue to form further back than normal (glossoptosis), Leads to airway obstruction, feeding difficulties, High risk for cleft palate
 - 3. Syndrome** - a pattern of features known to have a unifying underlying cause.
 - Down syndrome (trisomy 21) - not all features need to be present to make the diagnosis and people with some of the features might not have the disorder

Congenital Dysmorphology vocabulary:

4. **Disruption** - are due to destruction of a tissue or organ *after* it has been formed, a result of external forces and not a genetically controlled event
 - Amniotic band disruption, caused by fibrous **amniotic bands** while in utero - may result in finger amputation, or fusion of digits
5. **Deformation** – Mechanical distortions by a physical force upon an otherwise normal structure, occurring while the tissue is forming
 - Clubfoot that is caused by lack of amniotic fluid or intra-uterine crowding.
6. **Dysplasia** - an abnormality in a tissue structure caused by abnormal cellular organization.
 - Disorders caused by *FGFR3* mutation such as achondroplasia which affects many aspects of skeletal structure.
7. **Association** - are where traits coincide more often than expected by chance. The traits are seemingly unconnected and not due to a single genetic cause.

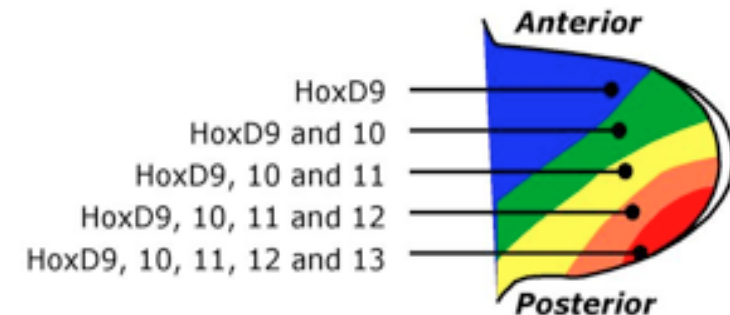
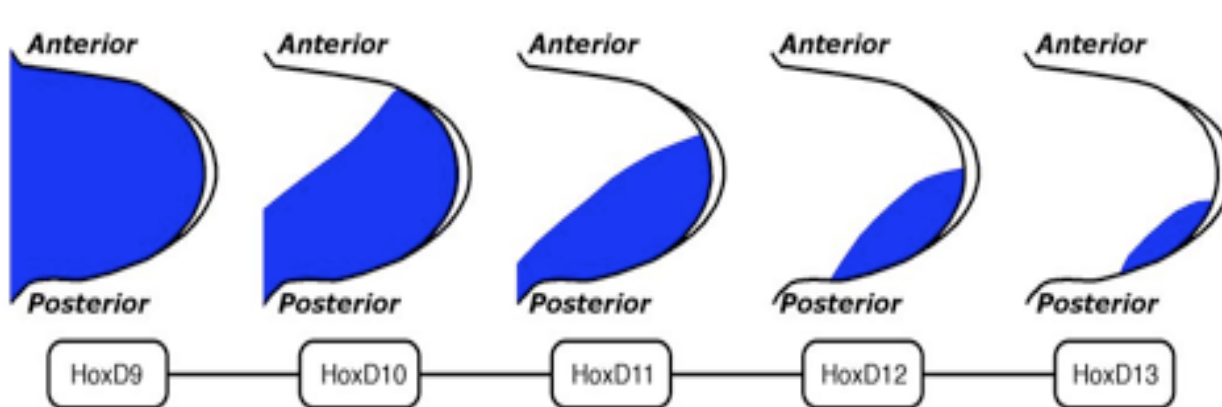
Sonic Hedgehog (SHH) Signaling Pathway:

- Sonic Hedgehog - soluble morphogen
- Induces genes involved in stimulation of cell proliferation and in tissue patterning.
- SHH protein cleaved to an active N-terminal form, which then binds to cholesterol.
- SHH/Chol binds to PTCH, removes lock on SMO function and activates downstream signaling.
- Stimulates GLI protein and CREBBP interact, **activating gene expression** by binding to regulatory elements in specific gene promoters.



SHH and Cell fate in development:

- SHH expressed in many tissues including:
 - Neural ectoderm - notochord and ventral floor plate of neural tube → DORSAL VENTRAL PATTERNING
 - Hair follicle, tooth, lung and pancreas development
- SHH expressed at different periods during development to control different areas
- SHH establishes the zone of polarizing activity (ZPA) - involved in limb patterning
- Pattern of Hox gene expression – responsible for setting anterior-posterior axis of the vertebrate embryo and the anterior-posterior axis of the vertebrate limb bud (established due to generation of a gradient of SHH protein produced by the ZPA).



Defects in the SHH Pathway:

- SHH mutation produce a range of different phenotypes.
- Heterozygous mutations leading to haploinsufficiency of *SHH* = **Holoprosencephaly** (loss of function mutations; Autosomal Dominant)
- Wide **clinical variability**, within same family: holoprosencephaly, microcephaly facial abnormalities including cyclopia, cleft lip/palate, hypotelorism (decreased interorbital distance), nose absent with proboscis in the middle of the forehead.
- Effect could be limited to single central maxillary incisor to no observable phenotypic observation at all = **variable expressivity to incomplete penetrance**



Defects in the SHH Pathway: Increased expression

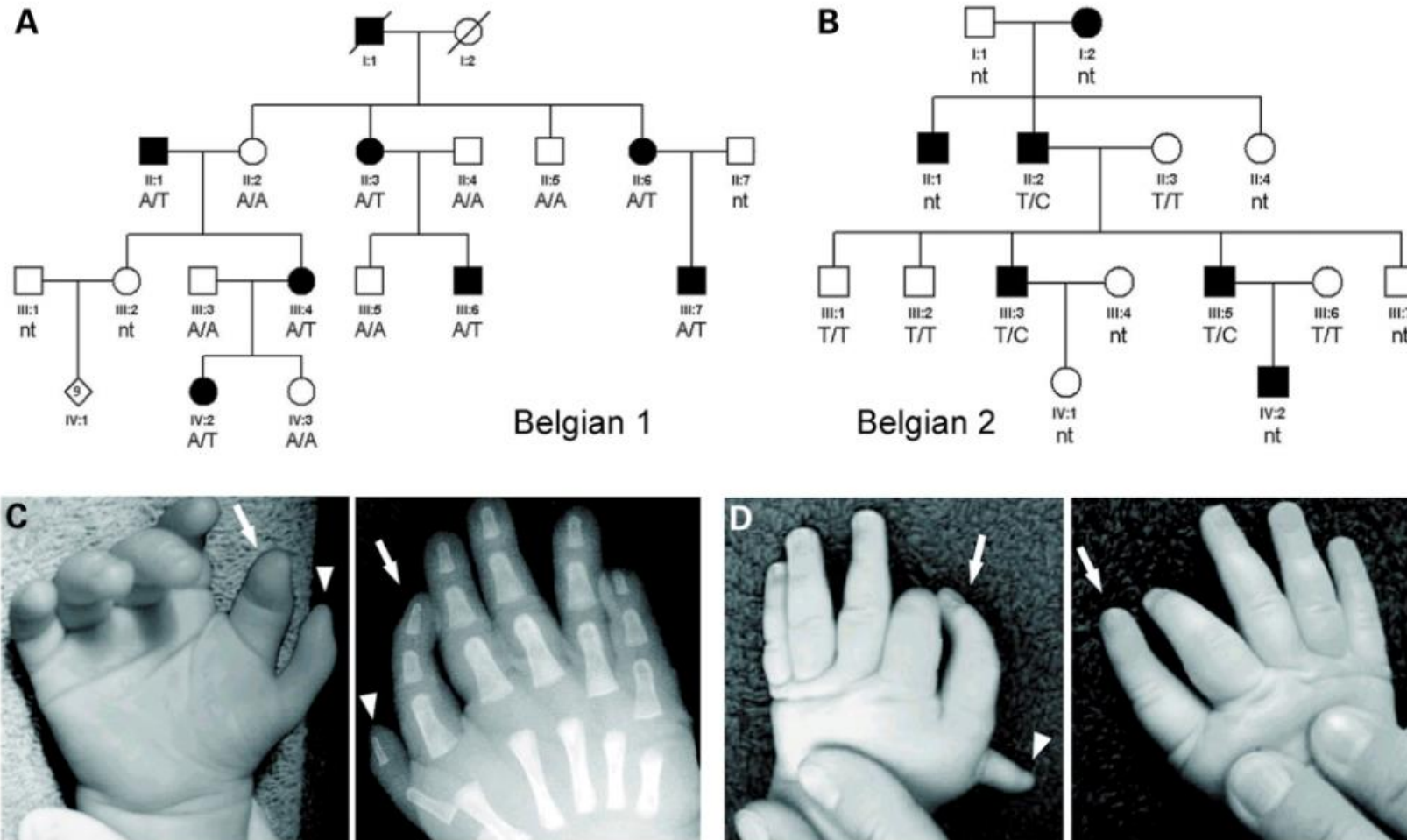
- Increased SHH expression in the frontal nasal prominence promotes mediolateral expansion of the face and ventral forebrain.
- Also increased distance between the eyes (**hypertelorism**)



- **True facial duplications extremely rare in humans** - Associated with other developmental disturbances that are incompatible with life.

Defects in the SHH Pathway: Increased expression

- **SHH** overexpression in the limb bud may be a cause of preaxial polydactyly.



Disease and Syndrome Spectrum of SHH Signaling Pathway

- **Holoprosencephaly (HPE)** (1/5,000 to 1/10,000 births)
 - One cause is haploinsufficiency of *SHH*
- **Phenotype/genotype correlation for *SHH* mutations difficult to establish**
 - Modifier effects of other genes
 - Spectrum of severity and variable expression
 - Mild - single incisor; Moderate - cleft lip and palate; Severe -cyclopia.
- **Smith-Lemli-Opitz Syndrome** (1/20,000 to 1/40,000 births)
 - Mutation in gene encoding 7-dehydrocholesterol reductase
 - Microcephaly, intellectual disability, malformation of mesodermal origin, syndactyly, and polydactyly
- **Other syndromes:**
 - Gorlin syndrome
 - Rubenstein-Taybi syndrome
 - Pallister-Hall syndrome
 - Greig cephalopolysyndactyly syndrome.

Summary and Genetic Principles Illustrated:

Terminology:

- Understand your terms clear and be able to give an example.

Sonic Hedgehog (SHH):

- Autosomal dominant – loss of function mutations
- Haploinsufficiency
- Variable expressivity
- Incomplete penetrance
- Locus heterogeneity – signal pathway

Thank you for listening!